KOMMU BILLY PRAISE PAUL

COMPUTER SCIENCE ENGINEER

+91 94916 58517 | billypraisepaulkommu@gmail.com | linkedin/billypraisepaulkommu | github.com/kommubillypraisepaul

SUMMARY

Computer Science Engineer with a strong foundation in machine learning, deep learning, artificial intelligence and data analysis. Proficient in Python, Java and SQL, with an understanding of AI model development, evaluation, optimization, and deployment. Strong problem-solving and analytical skills with a passion for learning emerging AI technologies and developing practical, scalable solutions.

EDUCATION

Vellore Institute of Technology – Amaravati **Sept 2022 - Aug 2026**

- B.Tech in Computer Science Engineering , CGPA: 8.35

Kendriya Vidyalaya No. 1, Vijayawada **Apr 2020 - June 2022**

- Higher Secondary (10+2), MPC (Maths, Physics, Chemistry) , CGPA: 8.46

Kendriya Vidyalaya No. 1, Vijayawada **Apr 2019 - June 2020**

- Secondary School (Class X) , CGPA: 8.37

SKILLS

Technical Skills

- **Programming Languages:** Java, Python
- **Web Development:** HTML, CSS, React.js, Node.js, Express.js
- **Cloud & DevOps:** AWS Cloud Practitioner Certified, Oracle Cloud Infrastructure (OCI) Generative AI
- **Databases:** SQL
- **Tools :** Canva , Blender

Soft Skills

Design Team Lead – Climate Change Club, VIT AP

- Led a 5-member design team in planning and executing campus-wide sustainability campaigns, creating impactful visual content and promotional materials to increase awareness and student participation.
- Collaborated with event and outreach teams to develop creative campaigns, posters, social media content, and event branding aligned with sustainability initiatives.
- Coordinated team responsibilities , Contributed to strengthening the club's engagement, visibility, and outreach through strategic visual storytelling and creative event collaborations.

WORK EXPERIENCE

MERN Full-Stack Developer Intern – Ethnus **June 2024 - July 2024**

- Developed a full-stack online quiz application using MongoDB, Express.js, React.js, and Node.js with role-based features for students and admins.
- Implemented REST APIs, authentication, quiz management, dynamic questions, timer-based sessions, and automatic scoring.
- Designed a responsive, interactive UI with smooth quiz navigation, result display, and improved learning experience.

PROJECTS

Sentiment Analysis on IMDb Reviews

- **GitHub:github/Sentiment-Analysis-DL**
- Tech: Python, TensorFlow, CNN, LSTM, NLP — Developed a CNN-LSTM deep learning model to classify approximately 50,000 IMDb movie reviews into sentiment categories, implementing a complete pipeline covering text preprocessing, tokenization, sequence padding, model training, and performance evaluation, achieving 87.88% accuracy with a 0.36 loss.

RL Based HumanCentric Image Editing Model

- Developed an RL-based image editor capable of executing multi-step natural language instructions for both global and object-specific image edits.
- Integrated spaCy, Grounding DINO, and SAM for prompt parsing, object detection, and pixel-level segmentation, enabling precise localized editing based on user intent.
- Implemented a PPO-based agent with CLIP-driven reward shaping and Human-in-the-Loop feedback, using Correct Action Bonus and Action Penalty mechanisms to improve editing quality and reduce reward hacking.

AI-Powered Meeting Summarizer

- Developed an end-to-end Flask-based web application that automates meeting documentation by accepting audio recordings and generating full transcripts and structured meeting reports.
- Integrated OpenAI Whisper (ASR) for verbatim speech-to-text transcription and Google Gemini Pro with prompt engineering to generate Key Decisions, assigned Action Items, and Executive Summaries from meeting transcripts.
- Implemented asynchronous frontend processing with JavaScript, secure file handling, temporary file cleanup, and JSON-based API communication, with evaluation showing reliable transcription, actionable summaries, and explicit task-ownership extraction.

RESEARCH PUBLICATION

Predicting Threatened Habitats Using Logistic Regression (ICCCT 2024, IEEE Xplore)-Research

- **IEEE paper link :ieeexplore.ieee.org/document/10984101**
- Developed a logistic regression–based predictive model to classify habitats as threatened or non-threatened using environmental factors including air quality, water pollution, tree cover loss, CO_2 emissions, urban population, and temperature rise.
- Collected and integrated environmental data from Global Forest Watch, World Bank, WHO, and Kaggle, followed by missing-value treatment, categorical encoding, an 80/20 train-test split, and 5-fold cross-validation for model validation.
- Achieved 93.55% classification accuracy, with correlation analysis identifying key relationships between environmental stressors and habitat health, providing an interpretable early-warning framework for conservation planning and resource prioritization.

TrustAnchor: End-to-End, Scalable Layer-2 Blockchain Provenance for Pharmaceutical Supply Chains (SASIGD 2026, IEEE Xplore)-Research

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- Designed an offline-first pharmaceutical provenance architecture integrating Polygon Layer-2 blockchain, IPFS, RBAC smart contracts, W3C DID/Verifiable Credentials, and Ed25519 cryptographic verification to combat counterfeit medicines.
- Developed a hybrid on-chain/off-chain architecture that anchors compact IPFS CIDs on Polygon L2 while enabling QR/NFC-based offline verification through a Progressive Web App.
- Evaluated the system through simulations and testnet measurements, achieving 0.770 ms median offline verification latency, 99.4% on-chain storage reduction, and $0.0008 single-batch anchoring cost, with security tests detecting QR replay and metadata tampering.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner –** Amazon (2025)
- **Oracle Certified Generative AI Professional –** Oracle Cloud (2025)
- **MERN Full Stack Certification –** Ethnus (2024)
- **A1 Certificate in German Language** - Goethe- Institute (2020)